

Anchor Representation: 2024 vs. 2026

Overview

This report compares anchor representation between the 2024 and 2026 calibration datasets.

The purpose is to determine which anchors were sufficiently represented in each calibration period and which anchors were excluded because they did not meet the representation criterion.

For **both 2024 and 2026**, an anchor is considered sufficiently represented if it appears in at least **80% of the available calibration files**.

This analysis distinguishes between:

- anchors retained in both analyses;
- anchors retained in 2024 but not 2026;
- anchors retained in 2026 but not 2024;
- anchors that appeared in the raw data but were excluded from both analyses.

1. 2024 Calibration Files

1.1 Load 2024 Files

- Total number of available 2024 calibration files: 29

1.2 Read 2024 Anchor IDs

- IDs are immediately converted to char to be consistent with 2026 data

1.3 Apply the 80% Representation Criterion to 2024 Data

- ID must appear in at least 80% of the available 2024 calibration files: 23.2 of 29 files
- Number of retained 2024 anchors: 672

2. 2026 Calibration Files

2.1 Load 2026 Files

- Total number of available 2026 calibration files: 89

We impose a filtering technique to ensure robustness for our comparatively wider period of calibration data collection for 2026. A number of datasets of the 2026 calibration data contain significantly few IDs, and would inhibit our analysis by presenting many missing data points. The 2026 representation analysis therefore uses only files containing at least 500 IDs.

- Number of 2026 files retained: 80

2.2 Read 2026 Anchor IDs

2.3 Apply the 80% Representation Criterion to 2026 Data

- ID must appear in at least 80% of the usable 2026 calibration files: 64 of 80 files
- Number of retained 2026 anchors: 899

3. Anchor Retention

3.1 Anchors Retained in Both Years

Number retained in both: 535

3.2 Anchors Retained in 2024 but Not 2026

137 anchors met the 80% criterion in 2024 but did not meet it in 2026.

3.3 Anchors Retained in 2026 but Not 2024

364 met the 80% criterion in 2026 but did not meet it in 2024.

3.4 All Raw Anchors Observed in 2024

691 unique anchors appeared at least once in the original 2024 calibration files.

3.5 All Raw Anchors Observed in 2026

965 unique anchors appeared at least once in the usable 2026 calibration files.

3.6 Anchors Excluded from Both Analyses

An anchor is considered excluded from both analyses if:

- it appeared in the raw 2024 and/or 2026 data
- it did not meet the 80% criterion in 2024
- it did not meet the 80% criterion in 2026

Number excluded from both: 65

4. Excluded Anchors

The following table combines the representation information for every anchor observed in either year.

```
# A tibble: 1,101 x 12
  id    files_2024 total_files_2024 proportion_2024 retained_2024 files_2026
  <chr>    <int>         <int>         <dbl> <lgl>         <int>
1 6042      22          29          0.759 FALSE          NA
2 6120      22          29          0.759 FALSE          NA
3 6295      22          29          0.759 FALSE          NA
4 6430      14          29          0.483 FALSE          NA
5 6508      17          29          0.586 FALSE          NA
6 6886      20          29          0.690 FALSE          NA
7 6939      20          29          0.690 FALSE          NA
8 7089      17          29          0.586 FALSE          NA
9 7194       2          29          0.0690 FALSE          NA
10 7260      23          29          0.793 FALSE          NA
# i 1,091 more rows
# i 6 more variables: total_files_2026 <int>, proportion_2026 <dbl>,
#   retained_2026 <lgl>, present_2024 <lgl>, present_2026 <lgl>,
#   classification <chr>
```

17. Summary of Anchor Classification

```
# A tibble: 5 x 2
  classification      number_of_anchors
  <chr>              <int>
1 Retained in both      535
2 Retained in 2026 only  364
3 Retained in 2024 only  137
4 Excluded from both; 2026 only    53
5 Excluded from both; 2024 only    12
```

18. Summary Table for the Report

A more compact summary is useful for reporting the results.

```
# A tibble: 6 x 2
  category                                number_of_anchors
  <chr>                                <int>
1 Retained in both 2024 and 2026          535
2 Retained in 2024 only                   137
3 Retained in 2026 only                   364
4 Excluded from both; present in both raw datasets      0
5 Excluded from both; present in 2024 only             12
6 Excluded from both; present in 2026 only             53
```

19. Representation Percentages

The following calculates the percentage of all anchors observed in either raw dataset that fall into each category.

```
# A tibble: 6 x 3
  category                                number_of_anchors percentage_of_all_obs~1
  <chr>                                <int>                <dbl>
1 Retained in both 2024 and 2026          535                0.486
2 Retained in 2024 only                   137                0.124
3 Retained in 2026 only                   364                0.331
4 Excluded from both; present in both ~      0                0
5 Excluded from both; present in 2024 ~     12               0.0109
6 Excluded from both; present in 2026 ~     53               0.0481
# i abbreviated name: 1: percentage_of_all_observed
```

20. Compare 80% Representation Rates

This table summarizes the number of anchors observed in each year and the number that passed the 80% criterion.

```
# A tibble: 2 x 6
  year total_files eighty_percent_threshold raw_anchors retained_80pct
  <chr>      <int>          <dbl>      <int>      <int>
1 2024         29            24        691        672
2 2026         80            64       965        899
# i 1 more variable: proportion_retained <dbl>
```

21. Anchors Retained in One Year but Not the Other

This table is particularly useful for identifying changes in anchor availability between calibration campaigns.

```
# A tibble: 501 x 2
  id      classification
  <chr> <chr>
1 6025 Retained in 2024 only
2 6039 Retained in 2024 only
3 6043 Retained in 2024 only
4 6054 Retained in 2024 only
5 6072 Retained in 2024 only
6 6082 Retained in 2024 only
7 6087 Retained in 2024 only
8 6088 Retained in 2024 only
9 6092 Retained in 2024 only
10 6094 Retained in 2024 only
# i 491 more rows
```

22. Anchors Excluded from Both but Observed in Either Dataset

This is the group of anchors that may be particularly useful for investigating why anchors failed the representation criterion.

```
# A tibble: 65 x 8
  id    files_2024 total_files_2024 proportion_2024 files_2026 total_files_2026
  <chr>    <int>         <int>         <dbl>         <int>         <int>
1 6042         22           29         0.759           NA           NA
2 6120         22           29         0.759           NA           NA
3 6295         22           29         0.759           NA           NA
4 6430         14           29         0.483           NA           NA
5 6508         17           29         0.586           NA           NA
6 6886         20           29         0.690           NA           NA
7 6939         20           29         0.690           NA           NA
8 7089         17           29         0.586           NA           NA
9 7194          2           29         0.0690          NA           NA
10 7260         23           29         0.793           NA           NA
# i 55 more rows
# i 2 more variables: proportion_2026 <dbl>, classification <chr>
```

23. Most Poorly Represented Anchors

The following table identifies anchors with the lowest representation in either year.

This can help identify anchors that were particularly intermittent.

```
# A tibble: 50 x 5
  id    representation_2024 representation_2026 minimum_representation
  <chr>          <dbl>          <dbl>          <dbl>
1 6025              1              0              0
2 6039              1              0              0
3 6042             0.759              0              0
4 6043              1              0              0
5 6054              1              0              0
6 6072              1              0              0
7 6082              1              0              0
8 6087              1              0              0
```

```

9 6088          1          0          0
10 6092         1          0          0
# i 40 more rows
# i 1 more variable: classification <chr>

```

24. Export Results

The major results are written to CSV files so they can be examined outside of the PDF report.

25. Final Interpretation

The 80% representation criterion is applied independently to the two calibration periods.

An anchor retained in 2024 therefore appeared in at least 80% of the available 2024 calibration files, while an anchor retained in 2026 appeared in at least 80% of the usable 2026 calibration files.

The comparison separates three scientifically distinct groups:

1. **Stable/repeated anchors:** retained in both 2024 and 2026.
2. **Year-specific anchors:** retained in one calibration period but not the other.
3. **Intermittent anchors:** observed in the raw data but excluded because they failed the 80% representation criterion.

The final group can be particularly informative because failure to meet the 80% criterion may itself reflect changes in anchor availability, connectivity, geographic coverage, or other properties of the calibration system.

```

# A tibble: 6 x 2
  category                                number_of_anchors
  <chr>                                <int>
1 Retained in both 2024 and 2026          535
2 Retained in 2024 only                   137
3 Retained in 2026 only                   364
4 Excluded from both; present in both raw datasets      0
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```